

# Statistical and Computational Models

## Lecture 8: Interpreting binary outcome models

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# The fitted logit

At the end of the last slide deck we fit this model:

```
logit <- glm(y ~ oil + lgdp, data = countries, family = binomial,  
            control = glm.control(epsilon = 1e-10))  
round(summary(logit)$coefficients[, 1:2], 3)
```

|             | Estimate | Std. Error |
|-------------|----------|------------|
| (Intercept) | -15.329  | 2.343      |
| oil         | -1.560   | 0.429      |
| lgdp        | 1.851    | 0.273      |

Three questions remain:

1. How do we characterize uncertainty?
2. How should we interpret the coefficients?
3. How is this statistical learning?

# Roadmap

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Uncertainty

Interpretation

Classification

## The curvature of the log-likelihood

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Recall that a spikier  $\ell$  pins  $\hat{\beta}$  down more tightly, so uncertainty is given by its curvature.

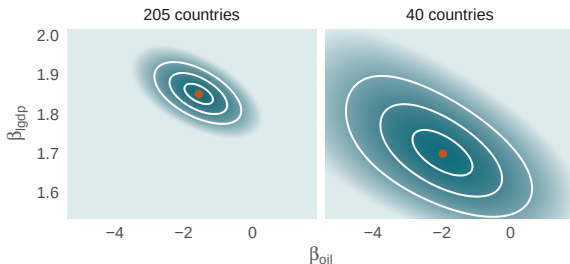
Curvature is the **Hessian** – the  $p \times p$  matrix of second derivatives, which tells us the **rate of change of the slope** of  $\ell$ .

$$H(\beta) = - \sum_{i=1}^n \pi_i(1 - \pi_i) \mathbf{x}_i \mathbf{x}_i^\top, \quad -H(\beta) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$$

$\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$ : fitted probabilities on the diagonal, with zeroes everywhere else.

Every weight  $\pi_i(1 - \pi_i)$  on the diagonal of  $\mathbf{W}$  is positive, so  $-H$  is positive definite and  $\ell$  is concave – its zero is the MLE.

## More information, less variance



Visualise  $\ell$  for different choices of  $\beta$ . The darker the colour, the closer to the **peak** shown in red.

The contour lines show  $\ell$  for a common fall from the peak – the tighter they are the spikier  $\ell$ . How far can  $\beta$  move before the fit gets worse by a set amount? With a flatter  $\ell$ , further –  $-H$  is smaller, so  $\left[-H(\hat{\beta})\right]^{-1}$  larger.

## From curvature to standard errors

We will omit the proof, but when  $n$  is large, the sampling distribution of  $\hat{\beta}$  itself is approximately normal:

$$\hat{\beta} \stackrel{\text{approx.}}{\sim} N\left(\beta, \left[-H(\hat{\beta})\right]^{-1}\right)$$

$-H(\hat{\beta})$  is the **observed information**: recall that at the peak  $\ell$  is concave, so  $H$  is negative, and the minus makes it positive.

$$\widehat{\text{Var}}(\hat{\beta}) = \left[\mathbf{X}^T \mathbf{W} \mathbf{X}\right]^{-1}$$

The variance of each  $\hat{\beta}_j$  down the diagonal, their covariances off it (think about the tilting rings in the previous slide). The standard errors are the square roots of that diagonal.

`summary()` reports these for any model fit by maximum likelihood. You are assuming the model is **true**, and  $n$  large.

## Standard errors in R

```
X    <- model.matrix(logit)
p    <- fitted(logit)
w    <- p * (1 - p)
V_eh <- solve(t(X) %*% diag(w) %*% X)
rbind(hand = sqrt(diag(V_eh)), glm = summary(logit)$coefficients[, 2])
```

|      | (Intercept) | oil       | lgdp      |
|------|-------------|-----------|-----------|
| hand | 2.342988    | 0.4289639 | 0.2727491 |
| glm  | 2.342988    | 0.4289639 | 0.2727491 |

Note: we must invert (`solve()`) the whole matrix in the last line.

General advice: if you doubt the logit form, `{sandwich}::vcovHC()`.

# Roadmap

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# Roadmap

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## References

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- Arel-Bundock, V. (2026) *Model to Meaning*. CRC Press – chs. 5–7. [marginaleffects.com](https://marginaleffects.com).
- Hanmer, M. J. & Kalkan, K. O. (2013) "Behind the Curve: Clarifying the Best Approach to Calculating Predicted Probabilities and Marginal Effects from Limited Dependent Variable Models." *American Journal of Political Science* 57(1): 263–277.